Confidential

Camera and LiDAR Based Pothole Detection and Reporting System
(Focus: Image Based Pothole Detection System)

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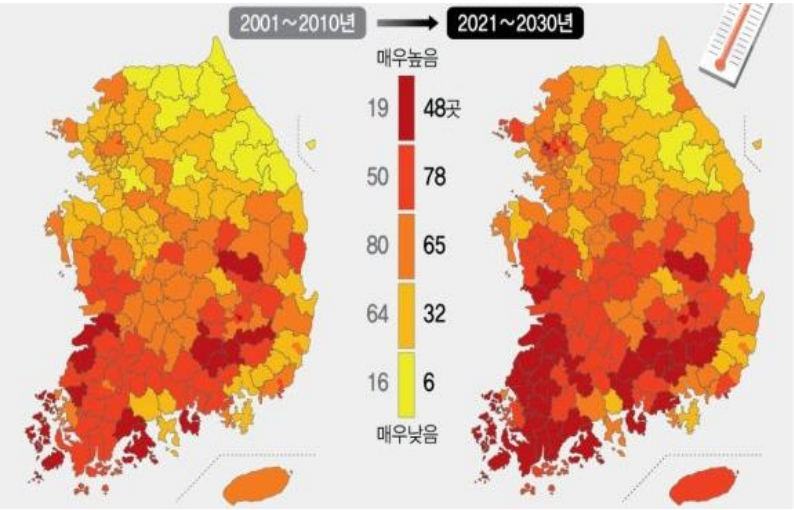
Introduction

Heavy snow

Heavy rain



Extreme heat

The frequency of potholes is increasing nationwide.

Introduction

Tire damage

Caused by potholes

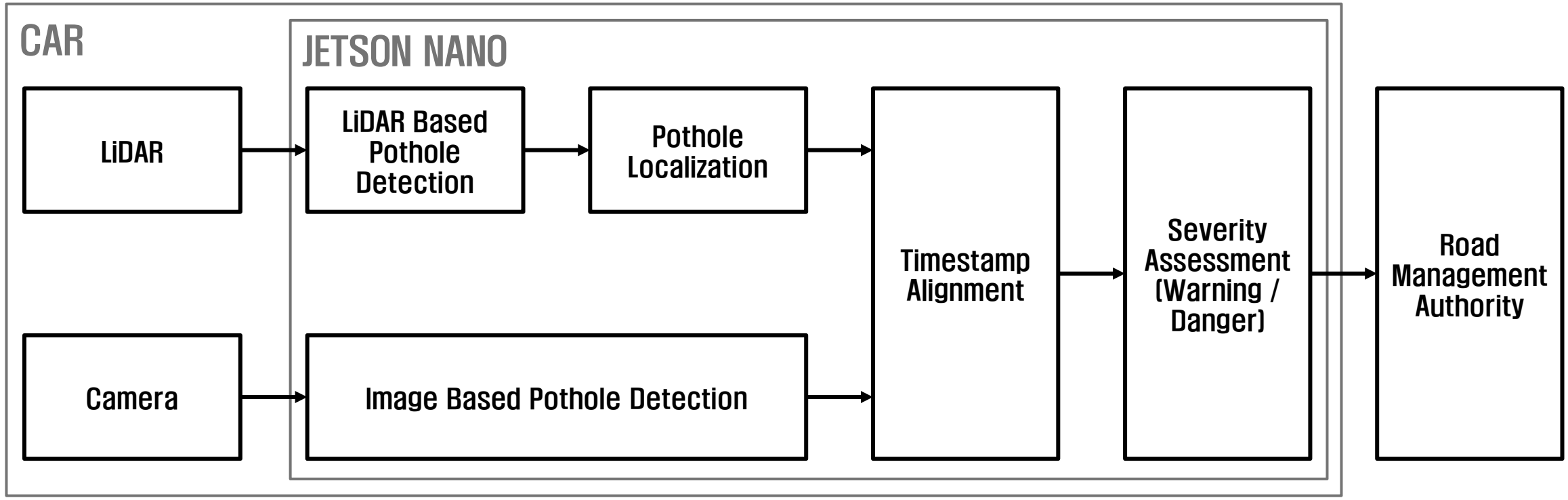
Traffic accident risk

Introduction

- **Problem**
  - Delayed repair
  - Low efficiency
  - No automatic assessment
- **Solution**
  - LiDAR & Camera fusion
  - Real-time
  - Automatic reporting
  - Multi-sensor reliability

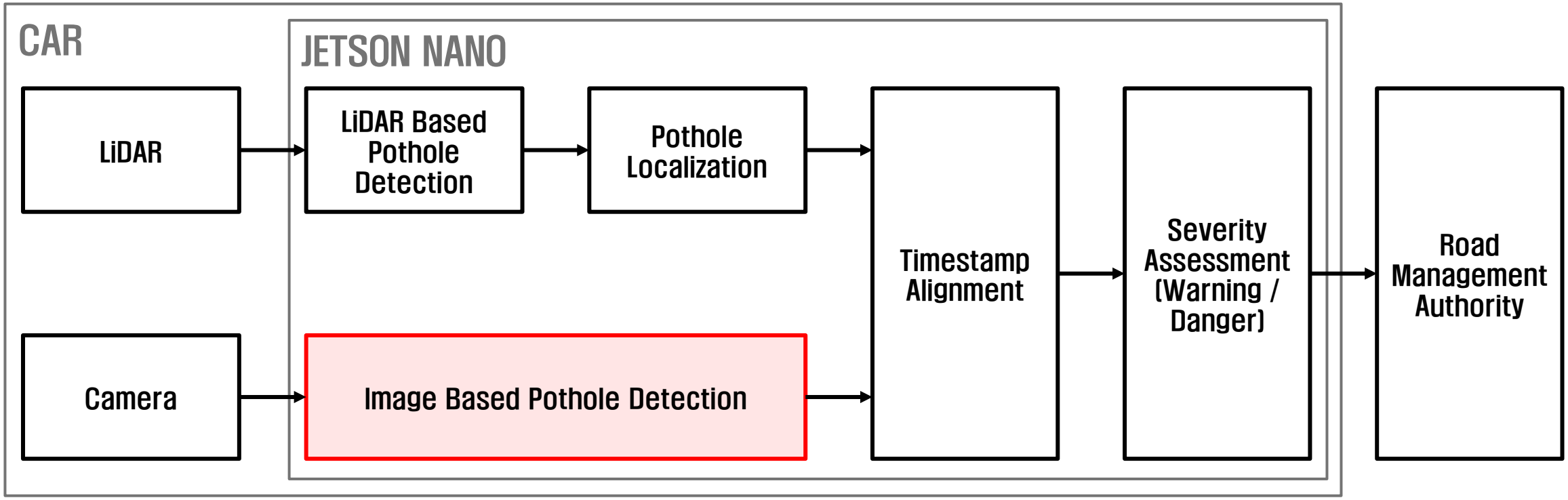
Proposed System

● Overall architecture



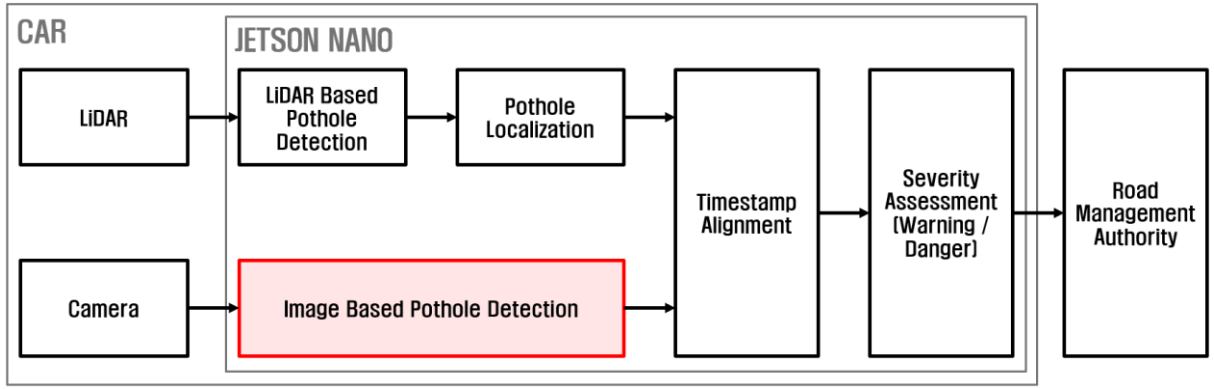
Proposed System

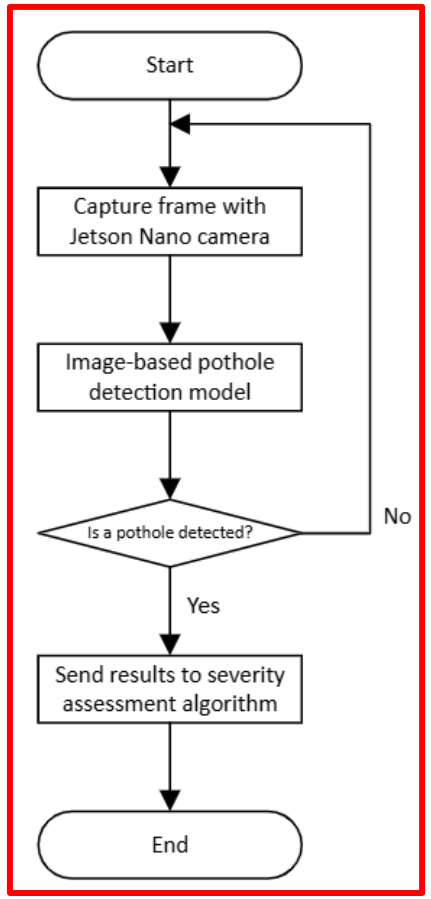
● Overall architecture



Proposed System

● Image Based Pothole Detection System





Methodology (Dataset preparation)

● Dataset Overview

Dataset	# of Images	Format	Preprocessing	Characteristics	Usage	image
1	15,445	YOLOv8	Resize (640×640)	Bottom-centered, biased	Partial use	
2	10,000	YOLOv8	Resize (640×640)	Centered, polygon labels	Fully used	
3	9,240	YOLOv8	Resize (640×640)	General roads, well-balanced	Main dataset	
4	655	YOLOv8	Resize (640×640)	Dense objects, small set	Supplement	
5	843	YOLOv4	Resize (640×640)	General roads	Only Test	

Methodology (Dataset preparation)

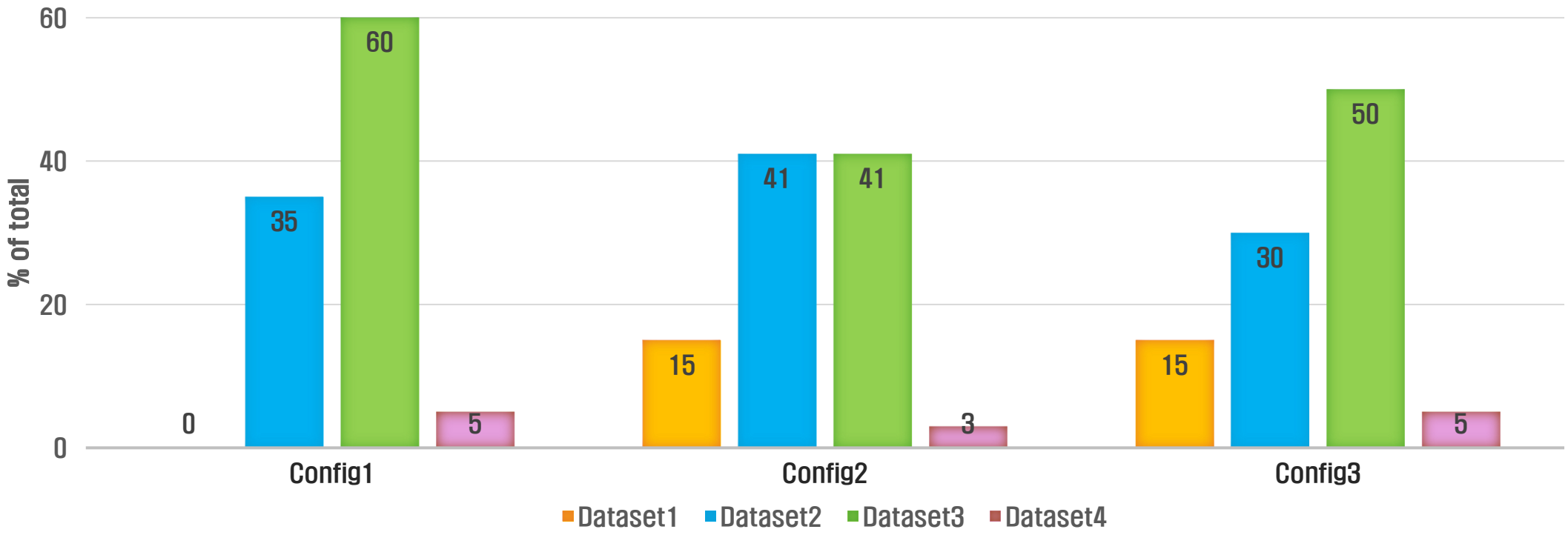
Data Cleaning

- Aspect ratio > 0.8
- Bounding box width > 0.8 (of image width)
- More than 5 boxes in a single image

⇒ All such objects were removed

Methodology (Dataset preparation)

● Dataset Composition by Configuration

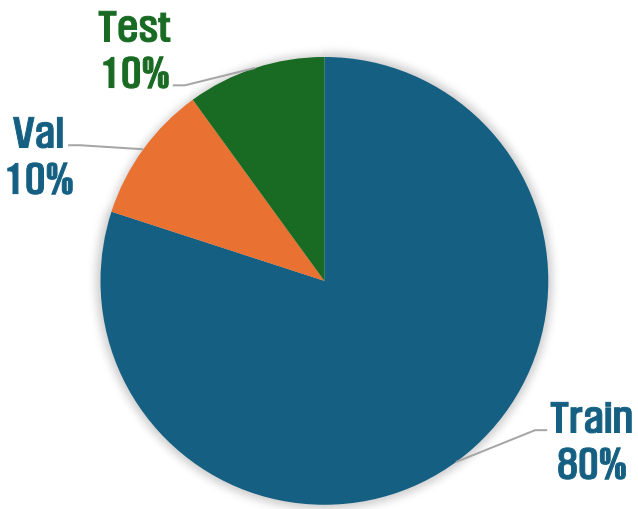


Total objects: 31,559 / 25,050 / 19,980

We tested three dataset configurations to find the best balance for training.

Methodology (Model fine-tuning)

● Dataset Split Ratio



● Hyperparameters

- Batch size: 32
- Epochs: 200
- Image size: 640 × 640
- Optimizer: AdamW, LR = 0.0005
- Early stopping: 10 epochs patience
- AMP & CosineLR: Enabled

● Training Environment

- Platform: Google Colab
- GPU: NVIDIA Tesla T4 (15GB VRAM)
- Python 3.11.13, Pytorch 2.6.0, Cuda 12.4

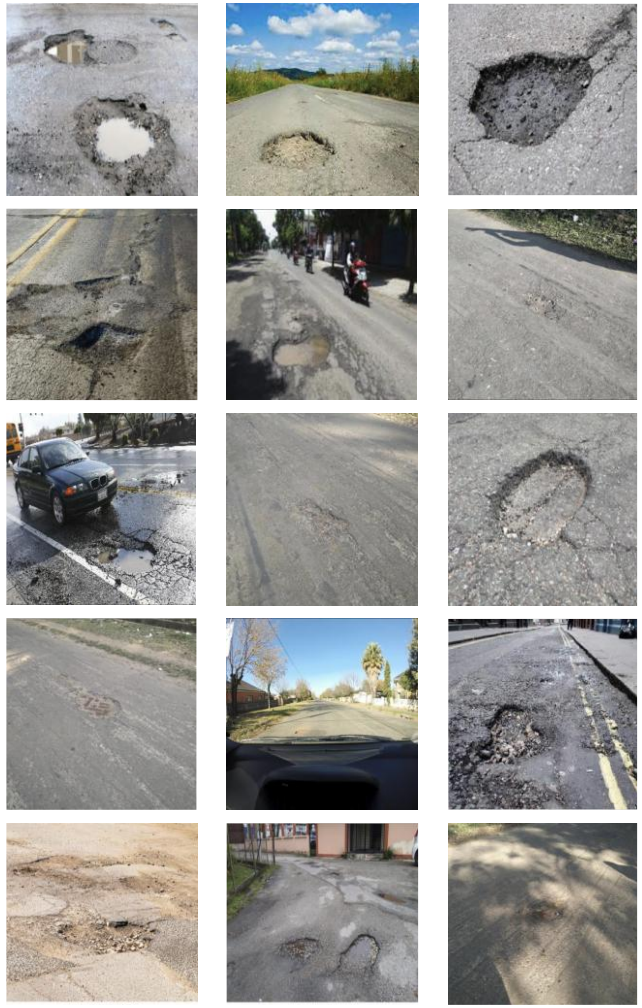
Results

Test Environment

- GPU: NVIDA Tesla T4 (VRAM: 15GB)

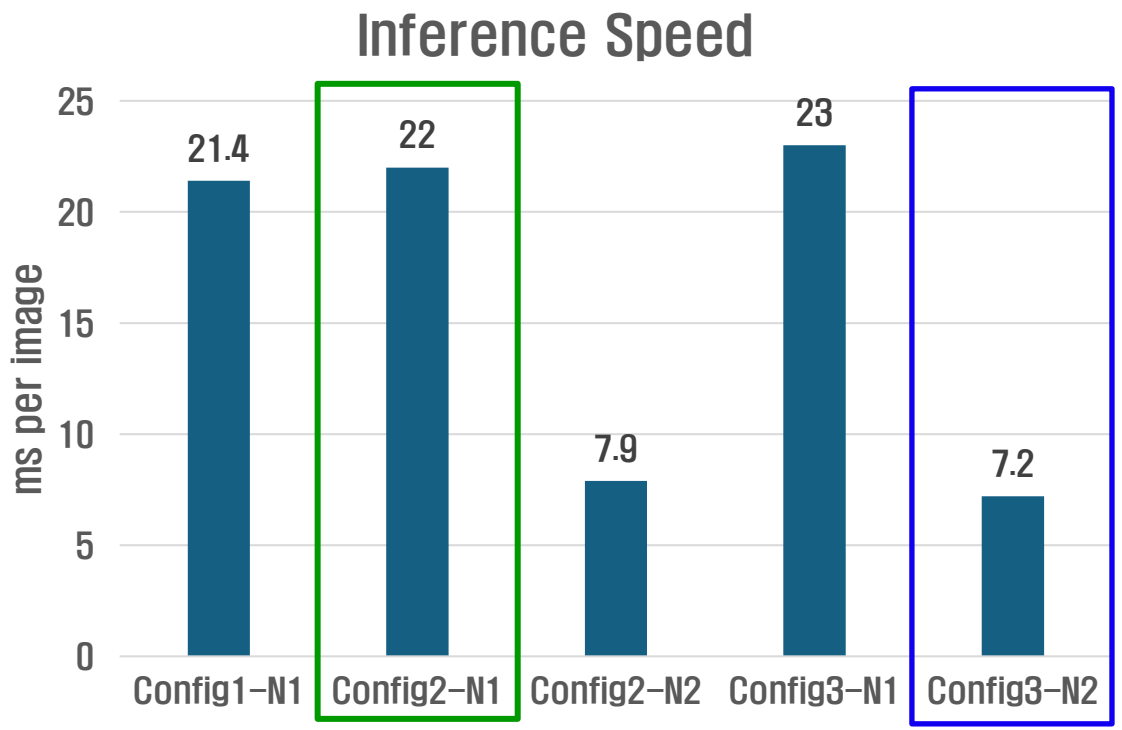
Evaluation Dataset

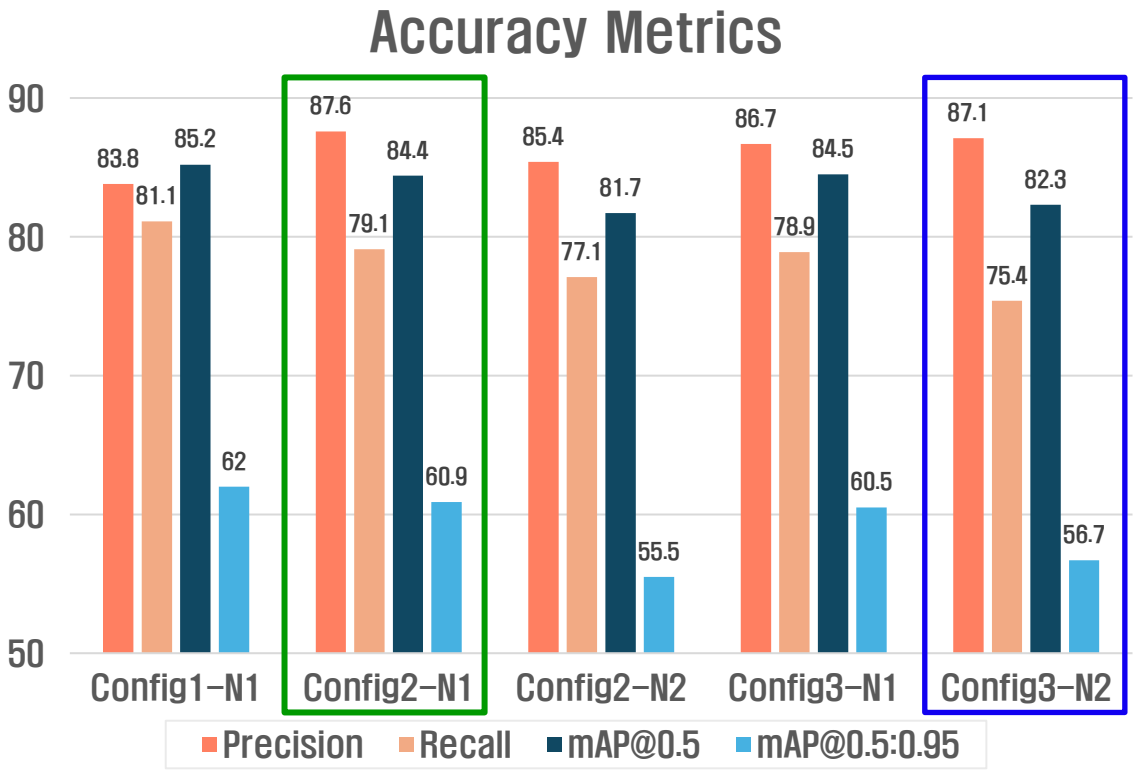
- Dataset 3 + Dataset 5 (Test sets)
- Well represents real road conditions



Results

● Performance Evaluation



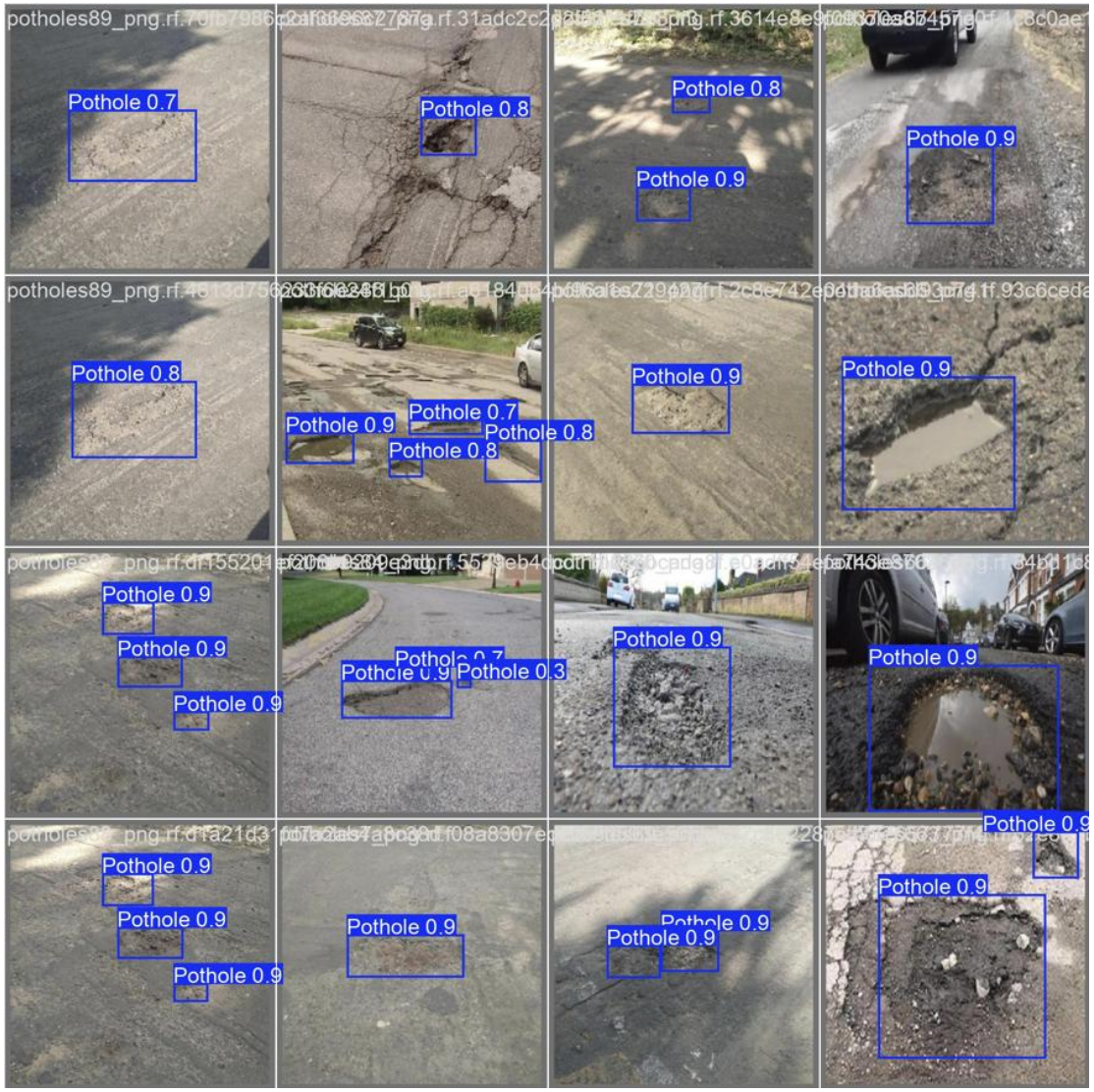


- Config3-N2 shows the fastest inference, but has lower recall.
- Config2-N1 provides the best balance of accuracy and speed.

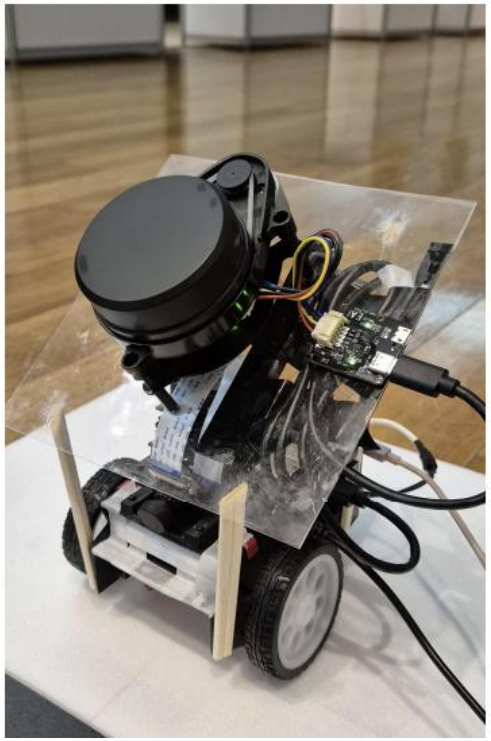
Results

Final Selected Model (Best Balance)

Config2-N1



Results

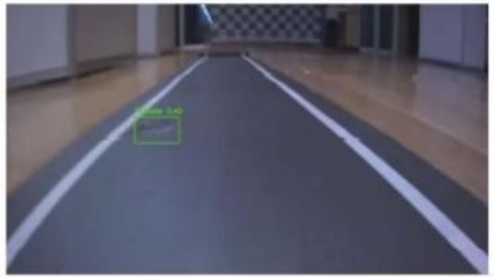


[실시간 포트홀 보고 로그]

시각: 2025-05-29 23:47:56.203

심각도: 주의

위치:
- 상대 거리: 15.38 m
- 시작: -
- 종료: -



시각: 2025-05-29 23:47:49.011

심각도: 주의

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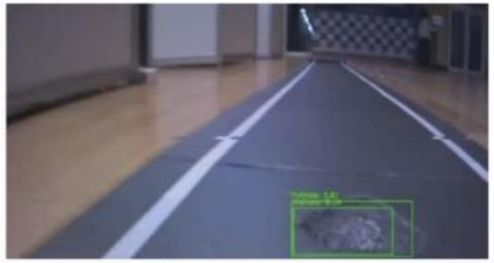
- 시작: -
- 종료: -



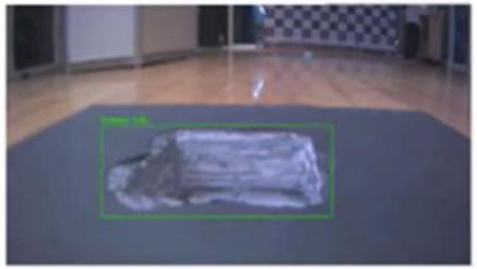
[실시간 포트홀 보고 로그]

시각: 2025-05-29 23:48:18.220

심각도: 위험

지름: 0.03 m

위치:
- 상대 거리: 19.87 m
- 시작: 0.35
- 종료: 0.32



시각: 2025-05-29 23:48:10.414

심각도: 주의

위치:
- 상대 거리: 18.28 m
- 시작: -
- 종료: -

Results

Demo Video

https://youtu.be/t9VONYIPrsI

Conclusion & Limitations

Conclusion

- Real-time (0.022s)
- Detection 79.1% (below target)
- Clear potholes → High Precision/Recall
- Feasible in practice

Limitations

- Detection rate gap
- Limited data diversity
- Duplicate detection issues
- Simple risk evaluation

Future Work

- Expand datasets
- Enhance post-processing
- Broaden detection scope
- Advanced risk assessment

References

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